

CUET PG (SCQP09) PRACTICE QUESTION BANK

Computer Science & Information Technology | Subject: Data Structures

TOPIC: STACKS (MEDIUM LEVEL)

50 MCQs

MAX MARKS: 200

Instructions:

- Each question carries 4 marks. For each incorrect response, 1 mark will be deducted (-1).
- Select the single best answer choice for each multiple-choice question.
- Answer Key is provided on the final page of this question sheet.

Q1. Which of the following permutation of numbers 1, 2, 3, 4, 5 CANNOT be obtained using a stack if numbers are pushed in order 1, 2, 3, 4, 5?

- (A) 4, 3, 2, 1, 5 (B) 1, 5, 4, 3, 2
(C) 3, 4, 5, 1, 2 (D) 2, 4, 3, 5, 1

Q2. Convert the infix expression $a + b * c + (d * e + f) * g$ into postfix form:

- (A) $abc*+de*f+g*+$ (B) $abc*+def*+g*+$
(C) $ab+c*de*f+g*+$ (D) $abc*de*f+g*++$

Q3. Evaluate the postfix expression: $6\ 2\ 3\ +\ *\ 4\ 5\ *\ +\ 9\ -$

- (A) 41 (B) 49
(C) 31 (D) 51

Q4. What is the time complexity to find the Minimum Element in a Min-Stack data structure using $O(1)$ auxiliary space beyond the stack?

- (A) $O(1)$ time and $O(N)$ extra space (B) $O(1)$ time using modified push/pop math
(C) $O(\log N)$ time (D) $O(N)$ time

Q5. When implementing two stacks in a single array of size N , what is the most space-efficient initialization for $top1$ and $top2$?

- (A) $top1 = 0, top2 = N/2$ (B) $top1 = -1, top2 = N$
(C) $top1 = -1, top2 = N - 1$ (D) $top1 = 0, top2 = N - 1$

Q6. Referring to Q5, what is the exact condition for Stack Overflow when pushing an element into either stack?

- (A) $top1 == N/2$ or $top2 == N$ (B) $top1 + 1 == top2$
(C) $top1 == top2$ (D) $top1 + top2 == N$

Q7. What is the postfix form of the infix expression $A \wedge B * C - D + E / F \wedge G$ (where $\wedge$ is exponentiation, right-associative)?

- (A) $AB^{\wedge}C*D-EFG^{\wedge}+$ (B) $AB^{\wedge}C*D-EFG^{\wedge}/$
(C) $AB^{\wedge}C*D-EF^{\wedge}G/+$ (D) $AB^{\wedge}C*D-+EFG^{\wedge}/$

Q8. What is the time complexity of finding the Next Greater Element (NGE) for all elements in an array using a Monotonic Stack?

- (A) $O(N^2)$ (B) $O(N \log N)$
(C) $O(N)$ (D) $O(2^N)$

Q9. In an algorithm computing the Maximum Rectangular Area in a Histogram of size N , what type of stack is maintained?

- (A) Monotonic Decreasing Stack of heights (B) Monotonic Increasing Stack of indices
(C) Min-Max Stack of pairs (D) Standard LIFO stack of raw values

Q10. What is the time complexity of calculating the Largest Rectangular Area in Histogram using a stack?

- (A) $O(N^2)$ (B) $O(N \log N)$
(C) $O(N)$ (D) $O(N^3)$

- Q11. To evaluate a Prefix expression efficiently using a stack, in which direction should the expression string be scanned?**
- (A) Left to Right (B) Right to Left
(C) From middle to both ends (D) Random order
- Q12. What is the prefix expression corresponding to infix $(A + B) * (C - D)$?**
- (A) $* + AB - CD$ (B) $* + A B - C D$
(C) $+ AB * - CD$ (D) $* AB + - CD$
- Q13. In implementing a Queue using 2 Stacks (s1 and s2), what is the amortized time complexity of the Dequeue operation if Enqueue is $O(1)$?**
- (A) $O(N)$ worst case, $O(1)$ amortized (B) $O(N)$ always
(C) $O(\log N)$ (D) $O(N^2)$
- Q14. In implementing a Stack using 2 Queues (q1 and q2) where Pop is $O(1)$, what is the time complexity of Push operation?**
- (A) $O(1)$ (B) $O(N)$
(C) $O(\log N)$ (D) $O(N^2)$
- Q15. An algorithm converts infix $A + B * C$ to postfix. What is the maximum number of operators inside the stack at any instance?**
- (A) 1 (B) 2
(C) 3 (D) 0
- Q16. What is the minimum number of stack frames required on the call stack to calculate $\text{Fibonacci}(5)$ using standard naive recursion?**
- (A) 5 (B) 6
(C) 15 (D) 9
- Q17. Which of the following is TRUE regarding stack-based recursion elimination?**
- (A) Tail recursion requires an explicit stack to eliminate (B) Non-tail recursion can always be converted using an explicit stack
(C) Recursion cannot be simulated using user-defined stacks (D) Stack elimination increases time complexity asymptotically
- Q18. Evaluate the prefix expression: $- * 2 3 + 4 5$**
- (A) -3 (B) 3
(C) 1 (D) -1
- Q19. Convert the prefix expression $+ * A B / C D$ into infix form:**
- (A) $(A * B) + (C / D)$ (B) $A * (B + C) / D$
(C) $(A + B) * (C / D)$ (D) $A + B * C / D$
- Q20. In standard expression evaluation, which statement accurately reflects operator associativity for exponentiation $^$?**
- (A) Left-to-right; incoming $^$ pops existing top $^$ (B) Right-to-left; incoming $^$ does NOT pop existing top $^$
(C) Non-associative; raises error on duplicate (D) Associativity depends on operand values
- Q21. How many stack pop operations are performed when converting $(A + B * (C - D)) / E$ into postfix format?**
- (A) 4 (B) 5
(C) 6 (D) 7
- Q22. What is the maximum height of the recursion call stack for Depth First Search (DFS) on a connected graph with V vertices and E edges?**
- (A) $O(E)$ (B) $O(V)$
(C) $O(V + E)$ (D) $O(\log V)$

- Q23. In the Stock Span Problem, where span $S[i]$ is defined as maximum consecutive days before day i with price $\leq$ price $[i]$, what data structure gives $O(N)$ total time?**
- (A) Queue of daily prices (B) Monotonic Stack storing indices of day prices
(C) Priority Queue (Max-Heap) (D) Binary Search Tree
- Q24. If array prices are $[100, 80, 60, 70, 60, 75, 85]$, what is the span array S using monotonic stack approach?**
- (A) $[1, 1, 1, 2, 1, 4, 6]$ (B) $[1, 2, 3, 4, 5, 6, 7]$
(C) $[1, 1, 1, 1, 1, 1, 1]$ (D) $[1, 2, 1, 2, 1, 4, 6]$
- Q25. What is the result of reversing a stack using recursion without using any extra auxiliary stack?**
- (A) Impossible without $O(N)$ extra memory (B) Achieved using Call Stack with insertAtBottom helper function
(C) Takes $O(N \log N)$ time (D) Modifies elements to garbage values
- Q26. What is the time complexity of reversing a stack recursively using the 'insertAtBottom' technique?**
- (A) $O(N)$ (B) $O(N^2)$
(C) $O(N \log N)$ (D) $O(2^N)$
- Q27. How many stacks are required to evaluate a complex infix expression directly in a single pass without converting to postfix first?**
- (A) 1 stack (B) 2 stacks (one for values, one for operators)
(C) 3 stacks (D) Cannot be done directly
- Q28. Convert the postfix expression $ab+cd-*$ to Infix form:**
- (A) $(a + b) * (c - d)$ (B) $a + b * c - d$
(C) $a + (b * c) - d$ (D) $((a + b) * c) - d$
- Q29. What is the output when pushing 10, 20, 30 into a Min-Stack, then popping 30, then calling getMin()?**
- (A) 30 (B) 20
(C) 10 (D) 0
- Q30. In K-Stacks implementation in a single array of size N using dynamic memory tracking, what extra arrays are maintained?**
- (A) top $[K]$ and next $[N]$ (B) top $[N]$ and next $[K]$
(C) head $[K]$ and tail $[K]$ (D) stack $[K][N]$
- Q31. Given the K-stacks structure in Q30, what is the space complexity of managing dynamic slot allocation?**
- (A) $O(N \times K)$ (B) $O(N + K)$
(C) $O(N^2)$ (D) $O(K^2)$
- Q32. In the Celebrity Problem (N people, 1 celebrity who knows no one but everyone knows them), how is a Stack used to solve it in $O(N)$ time?**
- (A) Push all persons, pop two at a time, eliminate one who knows the other (B) Store all known relationships in stack
(C) Sort elements on stack using quicksort (D) Pop elements until stack underflows
- Q33. What is the total number of matrix queries executed in the Celebrity Problem algorithm using stack elimination?**
- (A) $3(N - 1)$ (B) N^2
(C) $N \log N$ (D) 2^N
- Q34. What data structure combination is ideal for implementing an LRU (Least Recently Used) Cache?**
- (A) Stack and Array (B) Hash Map and Doubly Linked List
(C) Two Stacks and Priority Queue (D) Queue and Binary Search Tree

Q35. Trapping Rain Water problem can be solved in $O(N)$ time and $O(N)$ space using which stack technique?

- (A) Monotonic Decreasing Stack of bar indices
- (B) Monotonic Increasing Stack of water heights
- (C) Call Stack recursion
- (D) Two-way double ended queue

Q36. Given input sequence of operations on stack: `Push(1), Push(2), Pop, Push(3), Push(4), Pop, Pop, Pop`. What are popped values?

- (A) 2, 4, 3, 1
- (B) 1, 2, 3, 4
- (C) 2, 3, 4, 1
- (D) 4, 3, 2, 1

Q37. What is the auxiliary space complexity of sorting a stack using another temporary stack?

- (A) $O(1)$
- (B) $O(N)$
- (C) $O(N \log N)$
- (D) $O(N^2)$

Q38. What is the time complexity of sorting a stack using a temporary stack?

- (A) $O(N)$
- (B) $O(N^2)$
- (C) $O(N \log N)$
- (D) $O(2^N)$

Q39. Convert Prefix `+ - A B \* C D` to Postfix form:

- (A) $AB - CD * +$
- (B) $AB - C * D +$
- (C) $AB - CD + *$
- (D) $ABCD - * +$

Q40. In checking redundant brackets in an expression like `((a + b))`, what condition signals redundant brackets using stack?

- (A) When top element is `(` immediately upon encountering `)`
- (B) When stack contains no operators at all
- (C) When stack size exceeds $N/2$
- (D) When pop count equals push count

Q41. What is the result of simplifying a Unix path string `/a/./b/./../c/` using a stack?

- (A) `/c`
- (B) `/a/b/c`
- (C) `/a/c`
- (D) `/a/./c`

Q42. Decode an encoded string `3[a2[c]]` using two stacks (numberStack and stringStack). What is the expanded string?

- (A) accaccacc
- (B) acacac
- (C) aaaccc
- (D) accacc-c

Q43. What is the maximum number of distinct stack-permutations possible for an input sequence of N distinct elements?

- (A) $N!$
- (B) 2^N
- (C) $C_N = (1 / (N + 1)) * (2N! / (N! * N!))$ (Catalan Number)
- (D) N^N

Q44. For $N = 3$ elements (1, 2, 3), how many valid stack permutations can be generated?

- (A) 6
- (B) 5
- (C) 4
- (D) 3

Q45. In non-recursive Inorder Traversal of a Binary Tree, a stack stores:

- (A) Ancestor nodes whose left subtree is processed but right subtree is pending
- (B) Leaf nodes only
- (C) Nodes whose right subtrees are fully traversed
- (D) Child nodes before parents

Q46. In iterative Postorder Traversal of Binary Tree using 1 Stack, what condition requires inspection of `peek()`?

- (A) To check if right child exists and was not just visited
- (B) To balance tree height
- (C) To check if node is root
- (D) To swap left and right pointers

Q47. If an expression contains N operators and N+1 operands, how many total elements are pushed into operator stack during postfix conversion?

- (A) N (B) N + 1
(C) 2N + 1 (D) Depends on operator precedence

Q48. Which variant of stack allows element removal from both top and middle in constant time O(1)?

- (A) Standard Array Stack (B) Stack backed by Doubly Linked List with Mid-pointer
(C) Monotonic Stack (D) Circular Stack

Q49. In evaluation of postfix expression with stack, if an operator encounters less than 2 operands on pop:

- (A) Expression is invalid (Underflow error) (B) Operator is ignored
(C) Result evaluates to 0 (D) Operand default value 1 is used

Q50. In a Max-Stack supporting getMax() in O(1) time, pushing an element 'x' greater than current max 'M' updates stack with encoded value 'V' =:

- (A) $2x - M$ (B) $2M - x$
(C) $x + M$ (D) $x * M$

ANSWER KEY (Q1 - Q50)

Q.	Ans	Q.	Ans	Q.	Ans	Q.	Ans	Q.	Ans
1	C	11	B	21	A	31	B	41	A
2	A	12	A	22	B	32	A	42	A
3	A	13	A	23	B	33	A	43	C
4	B	14	B	24	A	34	B	44	B
5	B	15	B	25	B	35	A	45	A
6	B	16	B	26	B	36	A	46	A
7	A	17	B	27	B	37	B	47	A
8	C	18	A	28	A	38	B	48	B
9	B	19	A	29	C	39	A	49	A
10	C	20	B	30	A	40	A	50	A